Agisoft Metashape

Processing Report

08 September 2026



Survey Data

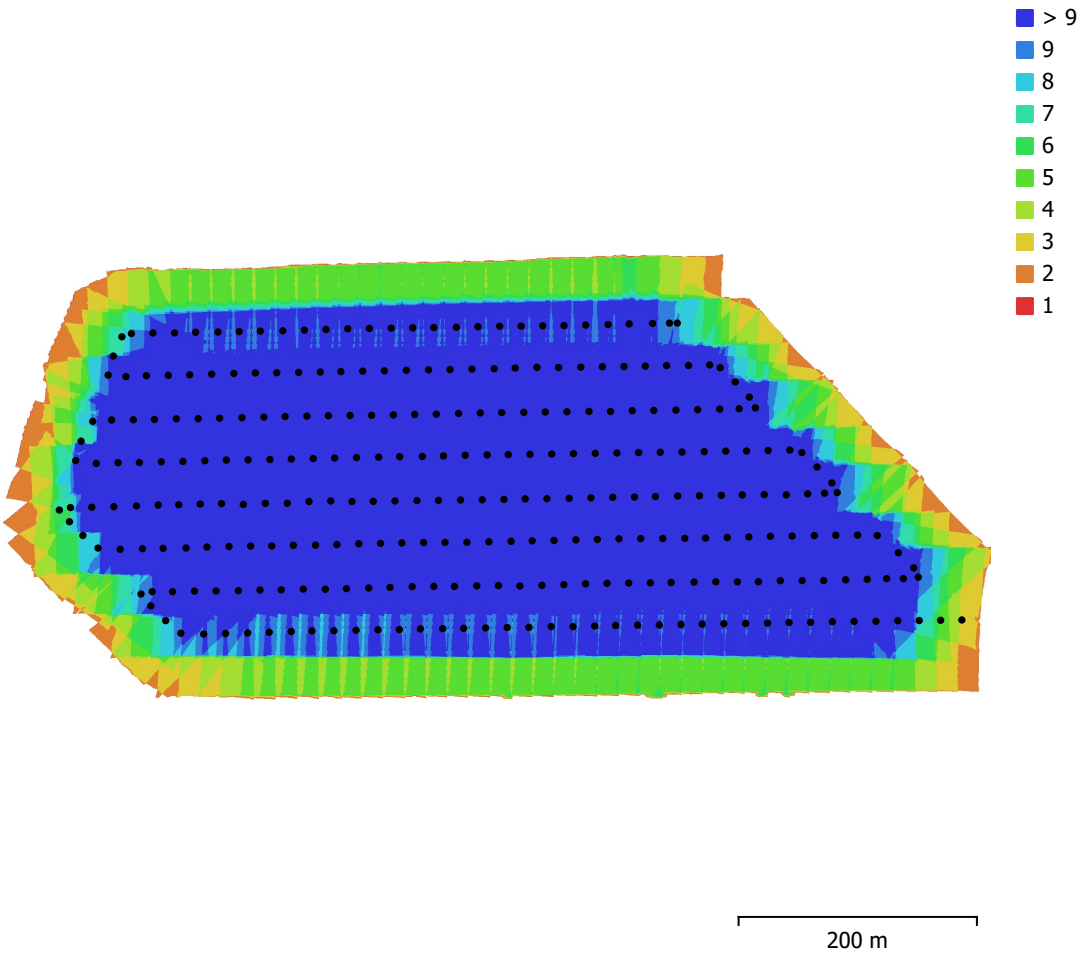


Fig. 1. Camera locations and image overlap.

Number of images:	287	Camera stations:	287
Flying altitude:	85.4 m	Tie points:	186,508
Ground resolution:	2.88 cm/pix	Projections:	661,655
Coverage area:	0.255 km²	Reprojection error:	0.656 pix

Camera Model	Resolution	Focal Length	Pixel Size	Precalibrated
ZH20T (4.5mm)	4056 x 3040	4.5 mm	1.6 x 1.6 µm	No

Table 1. Cameras.

Camera Calibration

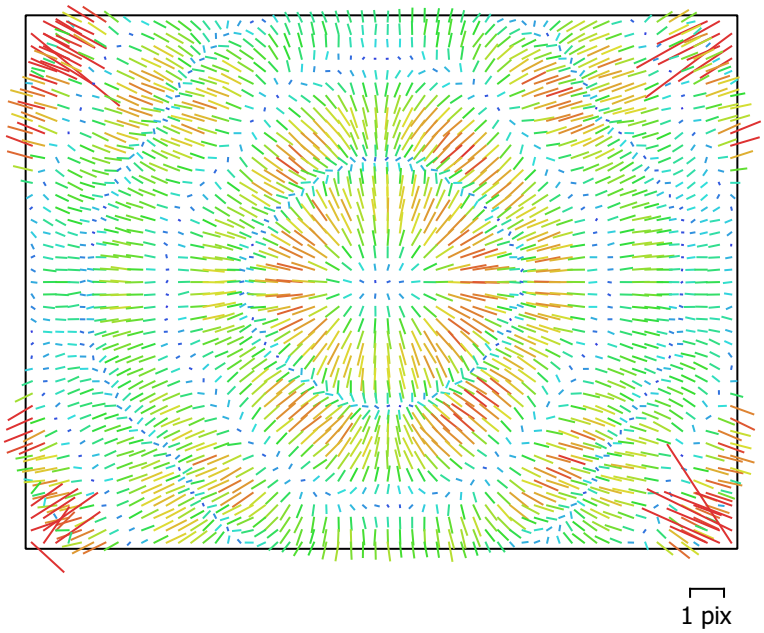


Fig. 2. Image residuals for ZH20T (4.5mm).

ZH20T (4.5mm)

287 images

Type	Resolution	Focal Length	Pixel Size
Frame	**4056 x 3040**	**4.5 mm**	**1.6 x 1.6 µm**

	Value	Error	Cx	Cy	B1	B2	K1	K2	K3	K4	P1	P2
F	**2811.66**											
Cx	**5.45395**	0.079	1.00	-0.06	0.07	-0.10	0.03	-0.02	0.02	-0.02	-0.59	0.03
Cy	**14.4185**	0.06		1.00	0.05	0.04	-0.01	0.01	-0.01	0.01	0.00	-0.24
B1	**-5.23181**	0.014			1.00	-0.02	0.01	-0.03	0.03	-0.02	-0.04	-0.07
B2	**0.324929**	0.014				1.00	-0.03	0.01	-0.01	0.01	0.08	-0.01
K1	**0.262908**	0.00015					1.00	-0.82	0.77	-0.76	-0.12	0.22
K2	**-0.87235**	0.00058						1.00	-0.99	0.96	0.03	-0.04
K3	**1.08297**	0.0011							1.00	-0.99	-0.02	0.03
K4	**-0.537857**	0.00076								1.00	0.03	-0.04
P1	**0.000234134**	4e-06									1.00	-0.06
P2	**-0.000589612**	4.2e-06										1.00

Table 2. Calibration coefficients and correlation matrix.

Camera Locations

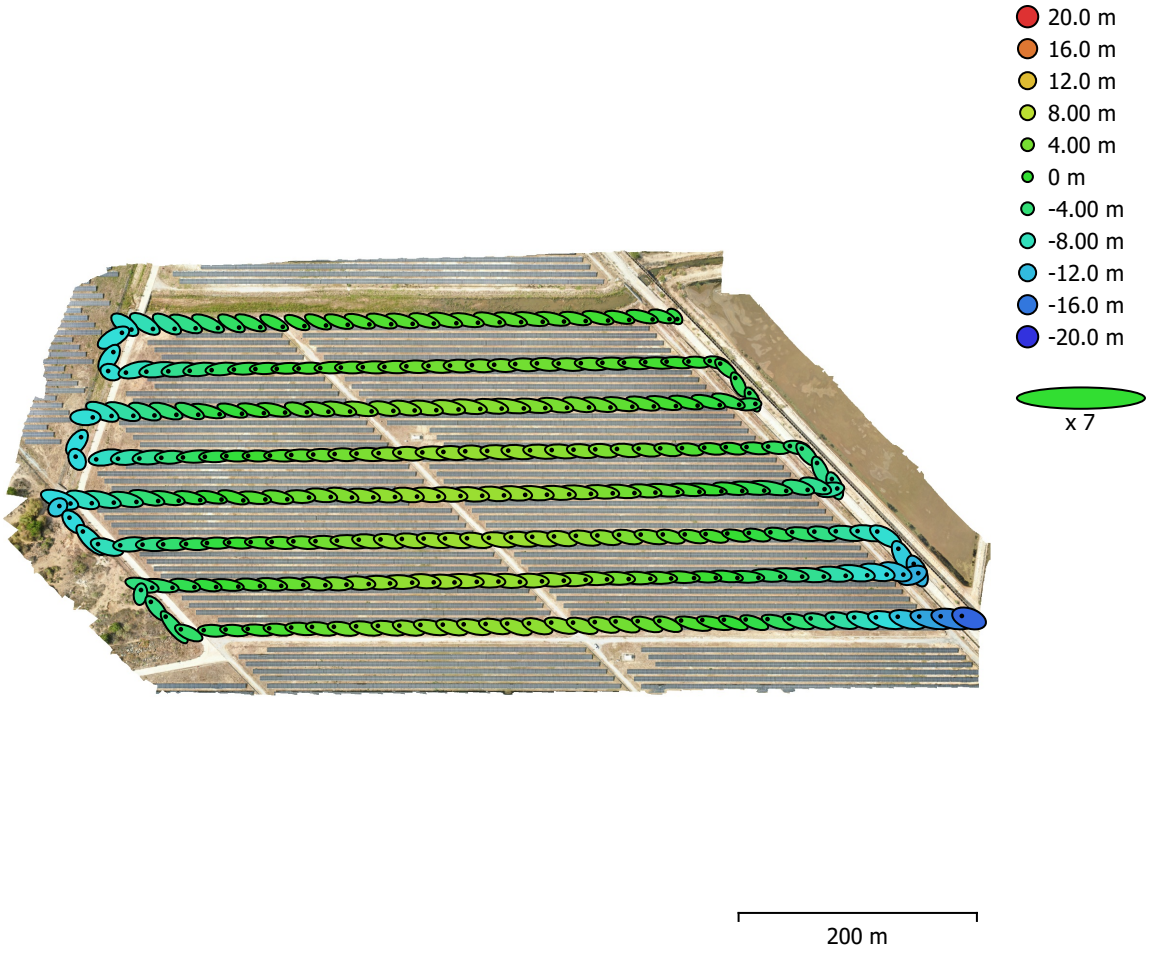


Fig. 3. Camera locations and error estimates.
Z error is represented by ellipse color. X,Y errors are represented by ellipse shape.
Estimated camera locations are marked with a black dot.

X error (m)	Y error (m)	Z error (m)	XY error (m)	Total error (m)
2.48879	0.644297	5.00607	2.57083	5.6276

Table 3. Average camera location error.
X - Longitude, Y - Latitude, Z - Altitude.

Digital Elevation Model

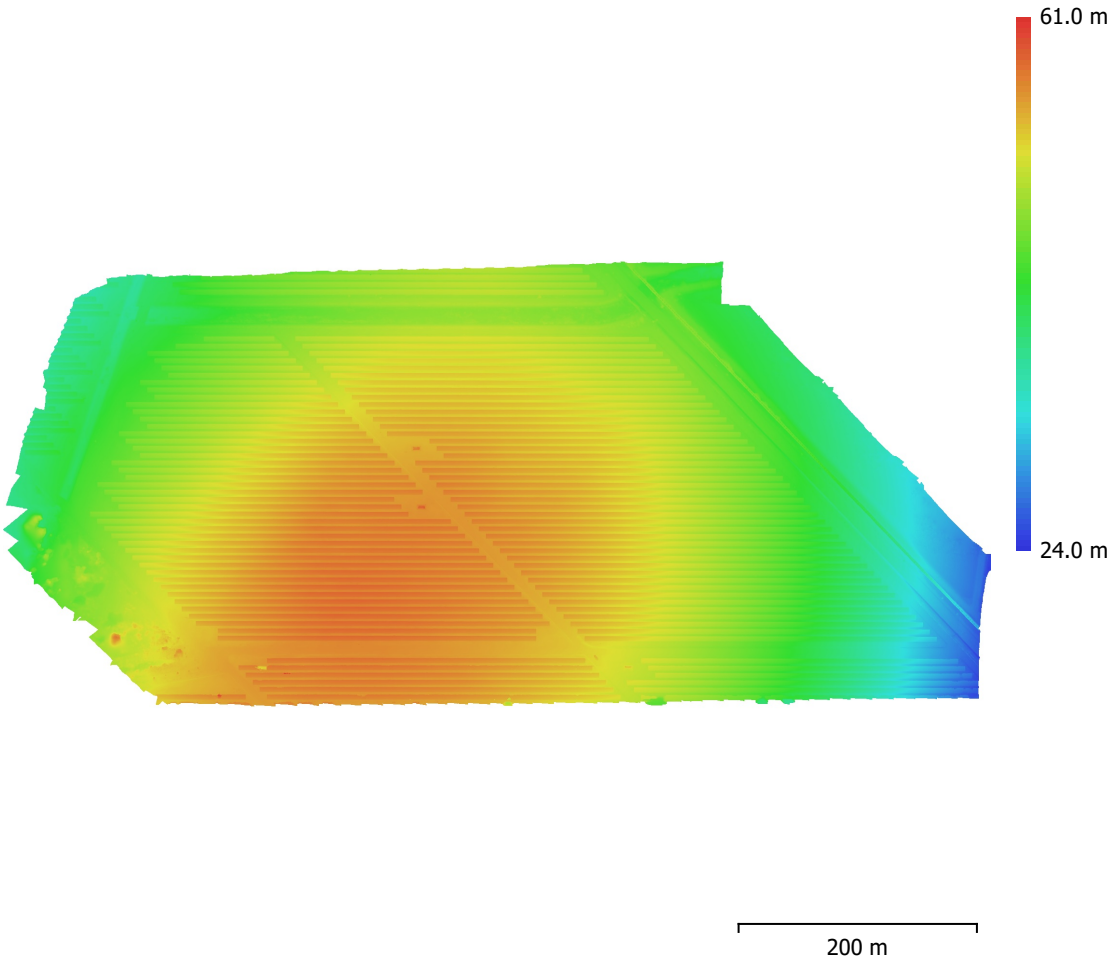


Fig. 4. Reconstructed digital elevation model.

Resolution:	6.07 cm/pix
Point density:	271 points/m²

Processing Parameters

General	
Images	287
Aligned images	287
Coordinate system	WGS 84 (EPSG::4326)
Rotation angles	Yaw, Pitch, Roll
Tie Points	
Points	186,508 of 349,200
RMS reprojection error	0.268286 (0.656378 pix)
Max reprojection error	1.46711 (4.85787 pix)
Mean key point size	2.49399 pix
Point colors	3 bands, uint8
Key points	943.74 MB
Average tie point multiplicity	3.59374
Alignment parameters	
Accuracy	High
Generic preselection	Yes
Reference preselection	Source
Key point limit	40,000
Key point limit per Mpx	1,000
Tie point limit	4,000
Exclude stationary tie points	Yes
Guided image matching	No
Adaptive camera model fitting	Yes
Matching time	10 minutes 1 seconds
Matching memory usage	402.20 MB
Alignment time	4 minutes 19 seconds
Alignment memory usage	320.04 MB
Optimization parameters	
Parameters	f, cx, cy, k1, k2, p1, p2
Adaptive camera model fitting	Yes
Exclude corners	No
Optimization time	8 seconds
Date created	2026:08:12 22:56:44
Software version	2.3.1.22580
File size	972.99 MB
Depth Maps	
Count	287
Depth maps generation parameters	
Quality	High
Filtering mode	Aggressive
Max neighbors	16
Processing time	14 minutes 21 seconds
Memory usage	2.72 GB
Date created	2026:08:12 23:23:05
Software version	2.3.1.22580
File size	1.40 GB
Point Cloud	
Points	89,499,201
Coordinate precision	1.44 cm
Point attributes	

Color	3 bands, uint8
Normal	
Point classes	
Created (never classified)	89,499,201
Depth maps generation parameters	
Quality	High
Filtering mode	Aggressive
Max neighbors	16
Processing time	14 minutes 21 seconds
Memory usage	2.72 GB
Point cloud generation parameters	
Source data	Depth maps
Processing time	20 minutes 52 seconds
Memory usage	4.82 GB
Date created	2026:08:12 23:44:01
Software version	2.3.1.22580
File size	1.17 GB
DEM	
Size	13,636 x 6,141
Resolution	6.07 cm/pix
Coordinate system	WGS 84 (EPSG::4326)
Depth maps generation parameters	
Quality	High
Filtering mode	Aggressive
Max neighbors	16
Processing time	14 minutes 21 seconds
Memory usage	2.72 GB
Reconstruction parameters	
Source data	Depth maps
Interpolation	Enabled
Processing time	41 minutes 22 seconds
Memory usage	6.59 GB
Date created	2026:08:13 01:12:39
Software version	2.3.1.22580
File size	306.69 MB
Orthomosaic	
Size	27,272 x 12,282
Resolution	3.04 cm/pix
Coordinate system	WGS 84 (EPSG::4326)
Colors	3 bands, uint8
Orthophotos	1.55 GB
Reconstruction parameters	
Blending mode	Mosaic
Surface	DEM
Enable hole filling	Yes
Enable ghosting filter	No
Processing time	4 minutes 45 seconds
Memory usage	1.40 GB
Date created	2026:08:13 01:15:01
Software version	2.3.1.22580
File size	1.97 GB
System	
Software name	Agisoft Metashape Professional
Software version	2.3.0 build 21954
OS	Windows 64 bit
RAM	15.78 GB

CPU	11th Gen Intel(R) Core(TM) i5-11400H @ 2.70GHz
GPU(s)	NVIDIA GeForce RTX 3050 Laptop GPU